DEEPAK MANOHARAN

AVIONICS ENGINEER | SOFTWARE DEVELOPMENT & VERIFICATION

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PROFESSIONAL SUMMARY

Avionics Software Engineer with 3.5 years of experience spanning helicopter avionics, embedded display software, and verification of safety-critical systems. Hands-on experience with Airbus Helicopters Helionix / Integrated Modular Avionics (IMA), Interface Requirement Specifications (IRS), signal-file generation and conversion workflows (KOJAK to generated C source), and PCF-related verification artifacts. Developed C/C++ display functionality for engine parameters and executed requirement-based LLT/HLT, integration, simulation, and hardware-in-the-loop testing. Working knowledge of DO-178B/C lifecycle practices, DOORS traceability, ARINC-429/653, build pipelines, and cross-team issue resolution.

ROLE-RELEVANT STRENGTHS

Helicopter avionics: Airbus Helicopters Helionix / IMA exposure

IRS and interface definition: Signal-to-display mapping; interface requirements

PCF and lifecycle artifacts: PCF verification; SVCP / SDD documentation

Verification and validation: Requirement-based tests; LLT/HLT; integration

Safety and standards: DO-178B/C; MIL-STD avionics processes

Data and integration: ARINC-429 / ARINC-653; simulator and target validation

PROFESSIONAL EXPERIENCE

CYIENT LIMITED | Verification & Validation Engineer ***/ Mar 2026 – Present***

- Develop requirement-based test scripts from DOORS requirements for Honeywell G300 aircraft avionics, implementing test logic against applicable MC/DC coverage objectives.
- Execute touchscreen controller (TSC) verification on mini-bench setups with the INAV module; validate cockpit display pages and touchscreen functions against software requirements.
- Simulate and validate TINK/ASCB data communication using TestProcess and customized TFM executables on mini-bench environments.
- Use Python to support test automation and defect analysis; execute and investigate test scenarios with TATS, TIU, ESCAPE, EASE, and MMT.

AIRBUS INDIA (VIA EXPLEO SOLUTIONS) | Software & System Engineer ***/ Apr 2025 – Dec 2025***

Project: Airbus Helicopters Helionix – Integrated Modular Avionics (IMA)

- Developed cockpit interface software **HELIONIX** and authored interface requirements, mapping system architecture signals to Multi-Function Display (MFD) and Aircraft Management Computer (AMC) inputs within an ARINC-653-based IMA environment.
- Worked on **IRS / GILDA** artifacts and delivered 20+ items in a two-month period; performed **PCF** verification, **SVCPR**-related checks, and supporting artifact verification.
- Created and maintained signal files and generated artifacts used to validate runtime behavior in simulation and on target hardware.
- Integrated signal artifacts into Git/Jenkins build workflows, maintained DOORS traceability, and collaborated with software and systems teams to resolve interface issues aligned with DO-178 lifecycle objectives.

EXPLEO SOLUTIONS, INDIA | Associate Software Engineer → Software Engineer **/ *Jun 2023 – Feb 2025***

Project: Electronic Engine Display (EED) / Multi-Function Display (MFD), Spanish F-5 aircraft

- Implemented C++ EED display features for real-time engine data including RPM, nozzle position, EGT, fuel flow, fuel quantity, and oil pressure; applied data filtering, unit conversion, and display formatting.
- Created 2,000+ test cases and scripts covering 550+ requirements; executed tests in simulation and hardware environments for an F-5 aircraft program.
- Performed ARINC-429 real-time testing using Mystic and WinTest and verified software behavior across operational modes using a Visual C++ 2015 cross-compiler workflow.
- Authored Software Design Documents (SDDs) and Software Verification Cases & Procedures (SVCPs); supported integration, instrumentation setup, ground simulation, and root-cause analysis of anomalies.
- Developed a traceability automation tool linking requirements across four key documents, reducing manual tracking effort; supported flight-test analysis and defect reporting.

TECHNICAL SKILLS

Avionics & protocols: Helionix / IMA, ARINC-429, ARINC-653, ARINC-717, Ethernet, RS-485, I2C, discrete I/O, analog signal interfaces

Development & verification: Embedded C/C++, Python, Java, IRS, signal-file generation, C-source generation workflows, LLT, HLT, MC/DC-driven test design, integration testing, simulation, HIL, defect analysis

Tools: IBM DOORS, Marcel Impact, KOJAK, GILDA, PCF, Git, Jenkins, Visual C++ 2015, Mystic, WinTest, Excaliber, TATS, TIU, ESCAPE, EASE, MMT, TestProcess, custom TFM executables, RTRT, Cantata

Standards & methods: DO-178B/C, MIL-STD-130, MIL-STD-461D, MIL-STD-704A, MIL-STD-1686A, MIL-L-85762A; Agile, V-Model

EDUCATION & CERTIFICATIONS

B.E. – Government College of Engineering (affiliated to Anna University), Salem, Tamil Nadu | Graduated 2023 | CGPA: 8.7

ISTQB Foundation Level | Concepts of RTOS – Udemy | Awards: Applause Award, Rising Star